OTC-06 | RELIABILITY & MAINTENANCE

Root Cause Analysis & Failure Investigation

SUGGESTED DURATION: 3 DAYS | IN-COMPANY / LIVE ONLINE

Move from repeated repairs to verified causes and effective corrective actions. Participants use a structured evidence trail to distinguish the physical failure mechanism from human and organisational contributors.

WHO SHOULD ATTEND

Maintenance, operations, quality and HSE teams; reliability engineers; supervisors and cross-functional investigation leaders.

WHAT PARTICIPANTS WILL BE ABLE TO DO

- Define a clear problem statement and preserve relevant evidence.
- Use timelines, cause-and-effect analysis, 5-Why and hypothesis testing appropriately.
- Select corrective actions with owners, completion evidence and effectiveness checks.

SUGGESTED PROGRAMME OUTLINE

Day 1 | Define and investigate

Failure event, scope and consequence; safe evidence preservation; operating history, interviews and physical evidence; timelines, barriers and problem definition; avoid blame and premature conclusions.

Day 2 | Analyse causes

Fishbone and 5-Why; physical mechanisms and contributing conditions; evidence-based causal chains; distinguish correlation from causation; test alternatives and document uncertainty.

Day 3 | Prevent recurrence

Corrective action hierarchy; feasibility, risk and management of change; action ownership and verification; effectiveness measures; communicating findings and applying lessons to similar assets.

OTC-06 | PRACTICAL APPLICATION & DELIVERY

Root Cause Analysis & Failure Investigation

PRACTICAL WORKSHOPS

Recurring bearing failure

Investigate an illustrative pump history with competing lubrication, alignment and operating explanations; identify evidence that would discriminate between them.

Investigation review panel

Groups present a causal chain and action plan; peers challenge unsupported assumptions and propose an effectiveness review.

PARTICIPANT TOOLS AND TAKEAWAYS

RCA problem statement; evidence register; cause validation worksheet; CAPA/action tracker; investigation report outline.

HOW LEARNING IS DELIVERED AND ASSESSED

Facilitated explanations, worked examples, case discussions and small-group exercises. A short baseline check establishes starting knowledge; a closing knowledge check and practical review assess learning. Assessment can be adapted for operators, technicians or experienced engineers.

CUSTOMISATION AND CLIENT INPUTS

A recurring plant problem, incident timeline and anonymised records. Client cases are used only with permission and appropriate confidentiality.

The agenda is indicative. Final content, duration, batch size, language and delivery arrangements are agreed after a training-needs discussion. Online delivery uses document reviews, calculations, role-play and case exercises; physical demonstrations require separate site arrangements.

FACILITATOR

Prem Prakash Mishra | Mechanical engineer, advisor and corporate trainer. Brings 39 years of career experience across Kuwait Oil Company, IPCL and NTPC, spanning projects, commissioning, maintenance, reliability, asset integrity and industrial audits. Specialist inputs, where needed, are agreed in the proposal.

COMPLETION AND FOLLOW-UP

An Omtech Corporation participation or completion certificate may be provided under the agreed assessment arrangements. This is company training, not an accredited professional qualification or a guarantee of certification. Post-training support is available with scope and duration agreed in the proposal.

REQUEST A CUSTOMISED TRAINING PROPOSAL

safehands@omtechcorporation.com

+91 9810772284 | +91 9987079111